

PROFILE

AI and Data Science undergraduate focused on building practical AI systems across **Indic NLP, retrieval-augmented generation, local LLMs, OCR, voice AI, and AI orchestration**. Experienced in taking projects from architecture and backend development through AI integration, testing, and deployment-oriented workflows. Strong interest in Tamil-first AI and production-minded software engineering.

TECHNICAL SKILLS

Programming	Python • TypeScript • SQL • JavaScript • HTML/CSS
AI / ML	NLP • RAG • Hugging Face Transformers • IndicTrans2 • Ollama • LoRA / Fine-tuning • Embeddings
Backend / Frontend	FastAPI • REST APIs • React • Vite • Tailwind CSS • WebSocket
Data / Retrieval	SQLite • PostgreSQL • ChromaDB • Vector Search • Semantic Search • BM25
AI Systems / Tools	OpenCV • TrOCR • faster-whisper • WebRTC VAD • Git • GitHub • Docker

SELECTED PROJECTS

TAMIL-AI.ME — Tamil-First NLP & RAG Platform

- Python • FastAPI • React • ChromaDB • IndicTrans2 • TrOCR • OpenCV
- Built a Tamil-first NLP/RAG platform for searching and working with Tamil literary content through a FastAPI backend and web interface.
 - Indexed 4,500+ literary entries across 22 sources, including Thirukkural and Sangam literature, using multilingual embeddings and semantic retrieval.
 - Integrated IndicTrans2 for Tamil–English translation and developed KalVettu, an OCR sub-module using TrOCR and OpenCV for digitizing Chola-era stone inscriptions.

SECRAIN — Local-First AI Memory & Reasoning Engine

- FastAPI • SQLAlchemy • ChromaDB • Ollama • RAG • Hybrid LLM Routing
- Designed a local-first AI data and memory system combining conversations, knowledge, semantic retrieval, and reasoning workflows.
 - Implemented automatic memory extraction, semantic search, notes / knowledge retrieval, and task-oriented data organization.
 - Built hybrid routing between cloud APIs and local Ollama models to support privacy-focused and offline-capable AI workflows.

TITAN — AI System Control Hub

- React • Vite • FastAPI • WebSocket • faster-whisper • WebRTC VAD
- Developed a personal AI assistant interface for voice control, task management, system interaction, and live workspace monitoring.
 - Implemented hybrid WebSocket and REST communication for real-time chat streaming and backend operations.
 - Integrated wake-word detection and speech processing with faster-whisper and WebRTC VAD for hands-free interaction.

ROGERS — AI Student Project Orchestrator

- FastAPI • React • TypeScript • Tailwind CSS • Local / Cloud LLMs
- Transforms one-line student ideas into structured, research-backed software projects with model selection, generation, testing, debugging, and delivery workflows.

AIRES — AI Emergency Response System

- React • FastAPI • PostgreSQL • AI Routing • Emergency Dispatch
- AI-powered emergency-response platform covering emergency intake, ambulance dispatch, route optimization, hospital recommendation, and simulated real-time traffic/CCTV/weather events.

EDUCATION

B.Tech — Artificial Intelligence and Data Science	2024–2028
Dhanalakshmi Srinivasan University, Perambalur, Tamil Nadu	
Higher Secondary (12th) — St. Joseph Academy, Sankarapuram	74%
Secondary School (10th) — St. Joseph Academy, Sankarapuram	88%

INTERNSHIPS

KaaShiv Infotech	Data Science Internship
IBM SkillsBuild AICTE Internship	via Edunet Foundation • Ongoing

CERTIFICATIONS

- Oracle Fusion AI Agent Studio — Certified Foundations Associate
- Tata Forage — GenAI Data Analytics
- Claude MCP Certificate

CORE STRENGTHS

AI Engineering	Building end-to-end AI applications rather than isolated model demos.
Indic NLP	Tamil-first language technology, multilingual retrieval, translation, and OCR.
Local-First AI	Practical use of Ollama and hybrid local/cloud model routing.
Product Engineering	FastAPI + React systems with APIs, retrieval layers, interfaces, and testing workflows.

PROJECT PORTFOLIO FOCUS

Tamil AI Ecosystem: TAMIL-AI.ME → SECRAIN → TITAN, with supporting systems such as ROGERS and AIRES. The portfolio demonstrates a progression from language intelligence and retrieval to persistent AI memory, voice interaction, system control, and multi-model orchestration.

CONTACT & LINKS

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